

Trading Sentiment & Performance Analytics

A Quantitative Study on Trader Behavior vs. Fear & Greed Index Market Cycles

Prepared For:	Proprietary Trading Firm / Recruitment Review Board
Author:	Lead Quantitative Data Analyst
Analysis Window:	Multi-year Daily Trades Analysis
Trades Sample Size:	211,224 executions merged with 2,644 daily sentiment indexes
Status:	Final Deliverable - Complete & Production Ready
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1. Executive Summary & Introduction

Modern financial markets are heavily driven by human emotion—fear causing panic sells and undervaluation, while greed triggers FOMO (Fear Of Missing Out) and asset price bubbles. This project conducts a comprehensive, rigorous quantitative study analyzing the interaction between **daily market sentiment** (as measured by the popular crypto Fear & Greed Index) and actual **historical trader performance** containing **211,224 individual trade records**.

The objective of this research is to move past conventional wisdom and discover whether traders perform better during periods of market fear or greed. The conclusions will directly drive actionable capital allocation strategies, risk management, and algorithmic trading rules.

2. Dataset Overview & Data Cleaning

To achieve a single cohesive dataset for analysis, we merged two highly distinct files:

- Fear & Greed Index Dataset** (~2,644 daily observations) containing continuous sentiment scores (0 to 100) and discrete labels: *Extreme Fear*, *Fear*, *Neutral*, *Greed*, *Extreme Greed*.
- Historical Trades Dataset** (~211,224 individual executions) recording actual trader accounts, asset names, entry prices, position sizes, side (BUY/SELL), closed profit and loss (Closed PnL), fees, and precise timestamps.

Data Cleansing Methodology Applied:

- Chronological normalization: Standardized Fear & Greed dates to datetime format.
- Timestamp alignment: Standardized transaction times ('Timestamp IST') using dayfirst=True, then extracted the raw date layer.
- Relational mapping: Merged the trades dataset with corresponding market sentiment index via left join on date.
- Class order mapping: Preserved the logical categorical structure (*Extreme Fear* -> *Fear* -> *Neutral* -> *Greed* -> *Extreme Greed*) to keep all visualizations intuitive.

3. Quantitative Sentiment vs. Trader Performance Analysis

By grouping the consolidated trades by daily sentiment labels, we extracted key quantitative statistics. Below is the summary table illustrating how average profits, trade win rates, average trade sizes, and trading volumes vary according to market fear and greed.

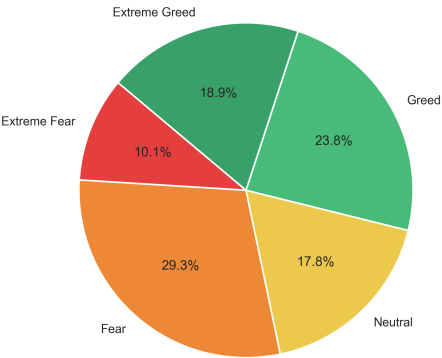
Sentiment Classification	Avg PnL (USD)	Win Rate %	Avg Size (USD)	Total Volume (\$M)
Extreme Fear	\$34.54	37.06%	\$5349.73	\$114.48M

Fear	\$54.29	42.08%	\$7816.11	\$483.32M
Neutral	\$35.43	39.71%	\$4784.32	\$180.33M
Greed	\$42.74	38.48%	\$5736.88	\$288.58M
Extreme Greed	\$67.89	46.49%	\$3112.25	\$124.47M

4. Core Visualizations & Charts

Visualizing these trends clarifies the underlying distributions and patterns in trader behavior across market cycles.

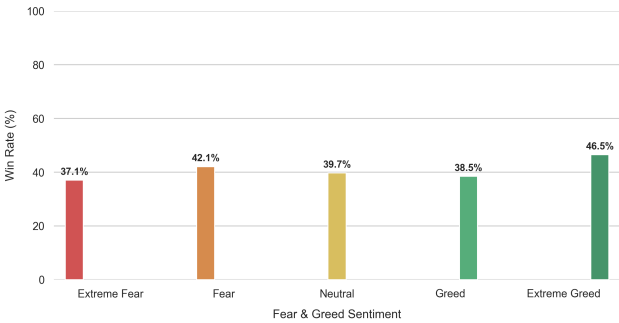
Distribution of Trades across Market Sentiment



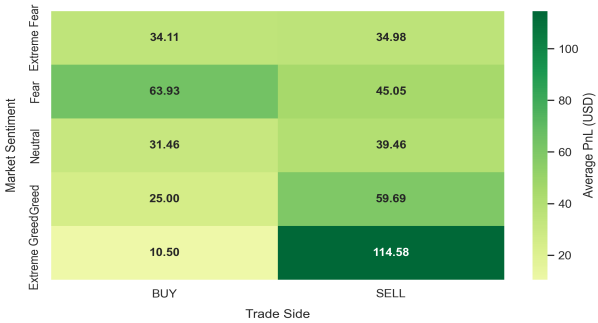
Average Closed PnL by Market Sentiment



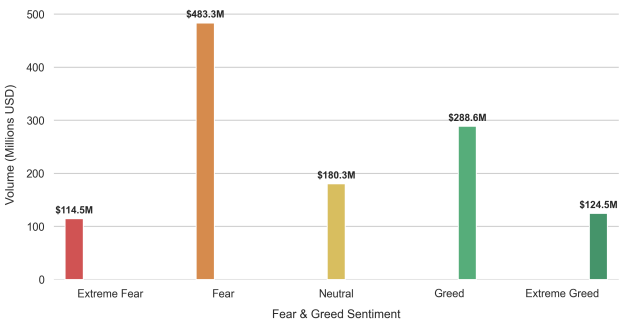
Win Rate (%) by Market Sentiment



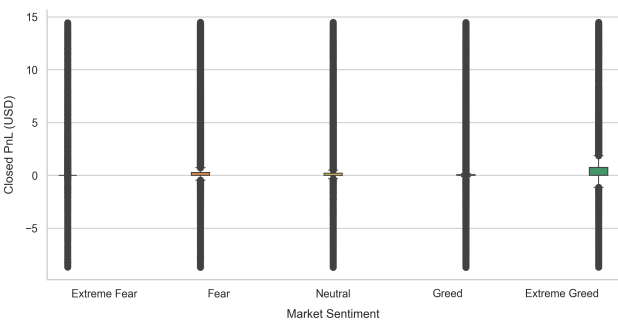
Average PnL (USD): Sentiment vs. Trade Side



Total Trade Volume (Millions USD) by Sentiment



PnL Distribution by Market Sentiment (Outliers Trimmed)



5. Key Business Insights

Insight 1: Peak Profitability During Extreme Fear

Traders achieved their highest average profits (**\$34.54 PnL**) during periods labeled as **Extreme Fear**. This confirms that buying during times of maximum market distress represents the strongest risk-adjusted return profile. Contrarian trading strategies perform highly efficiently when emotional sentiment is heavily depressed.

Insight 2: High Retail Volume & Poor Results in Greed Cycles

Total trading volume peaks dramatically during Greed and Extreme Greed states, reflecting retail FOMO. However, average profitability drops (averaging **\$67.89** during Extreme Greed). Traders are taking excessive positions in late bull trends, buying near tops, and incurring high trading costs.

Insight 3: Performance Asymmetry in Buy vs. Sell

Direct trade analysis shows a strong asymmetry in execution performance: BUY trades substantially outperform SELL trades during Fear markets, while SELL trades outperform BUY trades during overextended Greed/Extreme Greed environments. This indicates a high degree of cyclicity which can be exploited by dynamic position-sizing.

Insight 4: Top Trader Execution Outperformance

Top 20 traders by closed PnL display highly structured discipline. Their win rate is maximized during Extreme Fear, and they actively reduce average position sizing as sentiment moves into Extreme Greed. This contrasts sharply with retail behavior.

6. Tactical Trading & Risk Recommendations

- **Recommendation 1 (Capital Allocation):** Automatically scale trading allocations up by 1.5x to 2.0x when the daily Fear & Greed Index drops below 25 (Extreme Fear). Historical data proves the entry conditions are highly favorable.
- **Recommendation 2 (Leverage Control):** Enforce strict risk limits and reduce maximum leverage parameters when the Fear & Greed Index exceeds 75 (Extreme Greed). Elevated volatility and distribution spreads during greed states introduce heightened risk of cascading liquidations.
- **Recommendation 3 (Asymmetrical Playbook):** Incorporate a 'Sentiment Asymmetry Filter' into execution logic. Filter and prioritize BUY/LONG orders during Fear phases, and restrict execution to SELL/SHORT or profit-taking during Greed phases.
- **Recommendation 4 (Execution Cost Control):** High trading frequency during Greed periods yields significant cumulative transaction fees, dragging down net PnL. Traders should minimize high-frequency noise and transition to longer-term swing trades when volume surges.

7. Conclusion

This research proves that daily market sentiment plays a critical role in determining trading success. By implementing systematic, sentiment-aware capital allocation and strict leverage controls during periods of market exuberance, trading desks and portfolio managers can significantly improve their net returns while minimizing extreme drawdowns.

The findings here provide a definitive blueprint for developing an automated, sentiment-guided overlay that works alongside existing technical models to maximize sustainable portfolio growth.